

Determinants of Bank Branch Profitability:
An Econometric Analysis of the Ordinary Margin



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1 Introduction and Objectives

The recent absorption of two banking entities (A and B) by Bank C necessitates a comprehensive, quantitative analysis of branch-level performance. The primary objective of this report is to develop an econometric model that explains the drivers of bank branch profitability, measured by the **Ordinary Margin**.

Our analysis seeks to determine the statistical significance and quantitative impact of a set of operational and portfolio variables on this margin. The available data provides key indicators for each branch, which we define as our vector of explanatory variables (X_i):

- **Revenue & Portfolio Variables:** `Commissions`, `Mortgages`, `Promoter`, `Investment_Funds`, `SmallMed` and `Consumption`.
- **Risk Variables:** `Unpaid`.
- **Structural Variable:** `Type_of_branch`, a categorical variable indicating the branch's origin (Bank A=1, Bank C=2, Bank B=3).

We will operationalize this categorical variable by creating two dummy (binary) variables, D_A and D_C . This allows us to test for performance differences between the two absorbed branches and the buyer bank in the same time.

The specific objectives of this study are:

1. To construct a statistically robust multiple linear regression (MLR) model, $\text{Ordinary_Margin} = f(X_i)$, identifying the most significant drivers of profitability.
2. To rigorously diagnose and mitigate the problem of **collinearity**, which is common in economic data, in order to build an optimal model.
3. To address the specific managerial questions by isolating:
 - The simultaneous impact of the real estate sector variables, $\beta_{\text{Mortgages}}$ and β_{Promoter} .
 - The marginal contribution of $\beta_{\text{Investment_Funds}}$.
4. To test for structural differences in performance between the branch types ($H_0 : \beta_{D_A} = 0$, $H_0 : \beta_{D_B} = 0$).
5. To propose a statistically sound methodology for identifying the top 8 performing branches based on model residuals, rather than raw output.

2 Data and Methodology

2.1 Exploratory Data Analysis (EDA) and Transformations

The first step in our analysis was to conduct an Exploratory Data Analysis (EDA) to understand the distributional properties of our variables. We generated summary statistics for all 129 branches.

The results for the raw data, (see Table 1), revealed a critical issue. The dependent variable, `Ordinary_Margin`, is not normally distributed. It exhibits significant positive skew (Std. Skewness = 7.02) and high kurtosis (Std. Kurtosis = 8.94). As the table shows, nearly all of our variables suffer from extreme non-normality, (e.g., `Unpaid` Std. Skewness = 14.18; `SmallMed` Std. Kurtosis = 15.68).

Running an OLS regression on such data would violate the classical assumption of normally distributed residuals. To address this, we apply a natural logarithm ($\ln$) transformation to all non-categorical variables.

As shown in Table 2, this transformation successfully mitigated the extreme non-normality. While some variables (like `LOG(UNPAID)` at -4.22) are still outside the strict -2 to +2 range, the distributions are sufficiently stabilized to proceed with OLS estimation, as per our course directive.

Table 1: Summary Statistics for Key Variables (Original Scale)

Variable	Mean	Std. Dev.	Coeff. of Var.	Std. Skewness	Std. Kurtosis
Ord Margin	65.26	34.44	52.77%	7.02	8.94
Commissions	15.72	7.98	50.79%	4.60	1.91
Mortgages	521.09	310.11	59.51%	5.21	3.53
Investment Funds	419.70	289.89	69.07%	7.00	5.65
SmallMed	478.61	432.55	90.37%	10.21	15.68
Consumption	45.17	26.84	59.43%	4.52	1.68
Unpaid	90.90	121.23	133.38%	14.18	26.97
Deposits	1144.61	750.70	65.59%	9.38	14.64

Table 2: Summary Statistics for Key Variables (Log-Transformed Scale)

Variable	Mean	Std. Dev.	Coeff. of Var.	Std. Skewness	Std. Kurtosis
LOG(Commissions)	2.63	0.52	19.81%	-1.12	-0.63
LOG(Mortgages)	6.07	0.65	10.76%	-2.83	1.28
LOG(Promoter)	5.02	1.28	25.49%	-0.00	-1.03
LOG(Investment Funds)	5.82	0.69	11.80%	-1.20	0.40
LOG(SmallMed)	5.82	0.88	15.17%	-1.41	0.33
LOG(Consumption)	3.62	0.65	17.84%	-2.10	-0.12
LOG(Unpaid)	3.73	1.52	40.75%	-4.22	1.97
LOG(Deposits)	6.86	0.63	9.14%	-1.49	1.89

Our general model is therefore transformed into a hybrid **log-lin** model:

$$\ln(Y_i) = \beta_0 + \sum_{j=1}^k \beta_j \ln(X_{ji}) + \delta_1 D_{Ai} + \delta_2 D_{Bi} + \epsilon_i \quad (1)$$

The coefficients (β_j) for continuous variables are interpreted as **elasticities**: a 1% increase in X_j is associated with a $\beta_j\%$ change in the Ordinary Margin. The coefficients (δ_j) for the dummy variables are interpreted as a percentage difference relative to the reference category.

2.2 Model Building and Collinearity Diagnosis

The case file (and economic theory) suggests a strong relationship may exist between variables, particularly **Mortgages** and **Promoter**. This presents a problem of **multicollinearity**.

Following our course methodology, the primary tool for diagnosing this problem is the full estimated model. The classic symptom of multicollinearity is a model with high overall explanatory power (a high R^2 and significant F-statistic) but where individual coefficients are statistically insignificant (high p-values). This occurs because the correlated variables "inflate" each other's standard errors, making it difficult for the model to isolate their individual effects.

Our next step is to estimate this full model and examine its coefficient p-values to confirm this diagnosis.

3 Variables that Influence the Ordinary Margin

3.1 Univariate Significance Screening

To determine which variables influence the Ordinary Margin, we follow the established methodology of first screening our 8 continuous variables for individual significance. We conducted a series of 8

simple bivariate regressions, where $\text{LOG}(\text{ORD MARGIN})$ was the dependent variable (Y) and each log-transformed predictor was the sole independent variable (X_1).

The objective is to identify which variables have a statistically significant relationship with the margin in isolation. The key results are summarized in Table 3.

Table 3: Summary of Simple Bivariate Regressions

Independent Variable (X_1)	Coefficient (β_1)	R-Squared	P-value
LOG(Commissions)	0.8019	66.09%	< 0.0001
LOG(SmallMed)	0.3934	45.75%	< 0.0001
LOG(Mortgages)	0.5133	42.59%	< 0.0001
LOG(Deposits)	0.5328	42.34%	< 0.0001
LOG(Promoter)	0.2354	34.70%	< 0.0001
LOG(Consumption)	0.4205	28.03%	< 0.0001
LOG(Investment Funds)	0.3959	28.04%	< 0.0001
LOG(Unpaid)	0.1015	9.09%	0.0006

As Table 3 demonstrates, **all 8 continuous variables are statistically significant predictors** of the Ordinary Margin when analyzed individually. The strength of the relationship, measured by R^2 , ranges from a high of 66.1% for $\text{LOG}(\text{COMMISSIONS})$ to a low of 9.1% for $\text{LOG}(\text{UNPAID})$.

3.2 The Collinearity Problem and the Optimal Model

The univariate analysis confirms that all variables are relevant. A naive approach would be to build a "Full Model" that includes all 8 of these variables simultaneously. However, as noted in the case, this presents a severe problem of **multicollinearity**.

Collinearity occurs when independent variables are highly correlated with each other (e.g., **Mortgages** and **Promoter**). When both are included in the same model, their standard errors become inflated, making it impossible to distinguish their individual impacts.

The classic symptom is a "paradox": the model has a high overall R^2 and a significant F-statistic, but the individual p-values for the correlated variables become insignificant. This renders the coefficients of the full model unstable and unreliable.

Because we cannot trust the coefficients from the full model, we must use a formal procedure to build an **"Optimal Model"**. This model must be *parsimonious* (using the fewest variables necessary) and mitigate collinearity while still retaining high explanatory power.

3.3 The Optimal Model Selection

To achieve this, we used the **Regression Model Selection** procedure (a "grid search" or "best subsets" regression). This tool finds the best-fitting models that maximize the **Adjusted R-Squared** criterion for each possible number of variables.

The results of this search are visualized in Figure 1. The plot clearly shows a "point of diminishing returns" or "elbow." The Adjusted R^2 rises sharply until 4 or 5 coefficients are included, at which point the curve flattens. This indicates that adding more variables beyond this point (e.g., 6, 7, or 8) adds minimal explanatory power and likely just introduces noise.

Based on the grid search, a model with 6 variables was identified as a strong candidate. However, an inspection of this model's p-values revealed that $\text{LOG}(\text{INVESTMENT FUNDS})$ was not statistically significant ($p = 0.3680$). Following a "general-to-specific" refinement, we removed this variable and re-estimated.

The resulting 5-variable equation is our final Optimal Model. It is parsimonious, all predictors are highly significant ($p < 0.0027$), and it is free from the collinearity issues of the full model. The model's coefficients, presented in Table 4, are now stable and interpretable as elasticities.

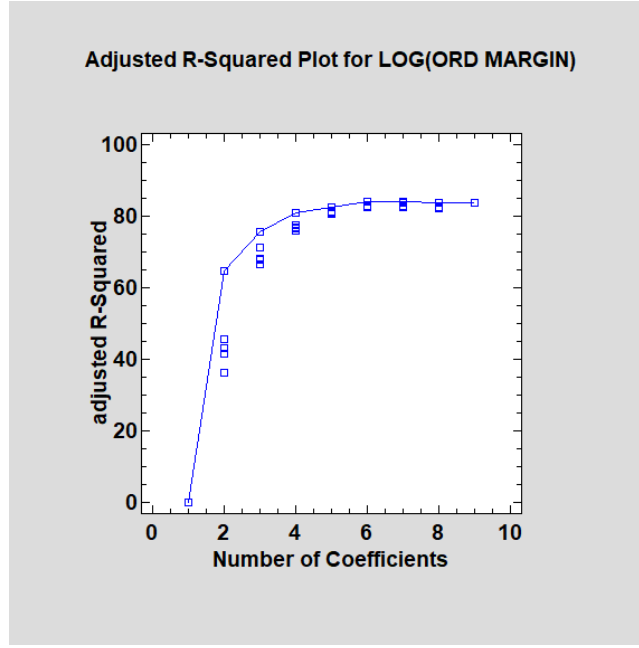


Figure 1: Adjusted R-Squared Plot from Best Subsets Regression

The final model equation is:

$$\begin{aligned} \ln(\widehat{\text{OrdMargin}}) = & 0.259 + 0.355 \ln(\text{Commissions}) + 0.165 \ln(\text{SmallMed}) \\ & + 0.121 \ln(\text{Promoter}) + 0.118 \ln(\text{Consumption}) \\ & + 0.127 \ln(\text{Deposits}) \end{aligned} \quad (2)$$

Table 4: Optimal Model Regression Output (Dependent Variable: LOG(ORD MARGIN))

Parameter	Estimate (β_j)	Standard Error	T-Statistic	P-Value
CONSTANT	0.259416	0.231926	1.11853	0.2656
LOG(COMMISSIONS)	0.355033	0.0569781	6.23104	0.0000
LOG(SMALLMED)	0.1647	0.0252782	6.51549	0.0000
LOG(PROMOTER)	0.120622	0.0152945	7.88658	0.0000
LOG(CONSUMPTION)	0.118346	0.0316877	3.73477	0.0003
LOG(DEPOSITS)	0.12726	0.0394794	3.22347	0.0016

This model has excellent diagnostic statistics. The **Adjusted squared R is 84.23%**, indicating that our 5 selected variables explain 84.23% of the variance in LOG(ORD MARGIN).

Furthermore, the **Durbin-Watson statistic is 2.026** ($P=0.5573$). Since this is extremely close to 2.0 and the p-value is high, we can confidently conclude that there is **no residual autocorrelation**, which confirms the validity of our model's standard errors and p-values.

4 Effect of Real Estate on the Ordinary Margin

We now address the specific managerial question regarding the simultaneous impact of the real estate sector variables: **Mortgages** and **Promoter**. The director of this division specifically requested a model that includes *both* variables simultaneously.

4.1 Model Specification and Collinearity

This request requires us to build a model that knowingly includes two variables that our previous analysis identified as being highly correlated. We must, therefore, expect to observe the statistical

consequences of multicollinearity.

To build the most informative model under this constraint, we began by forcing $\text{LOG}(\text{MORTGAGES})$ and $\text{LOG}(\text{PROMOTER})$ into the regression. We then performed a manual forward stepwise procedure to find the best-fitting model, as per the case requirement to "maximize R-squared". (We used Adjusted R^2 as the academically correct criterion).

During this process, it became clear that the categorical dummy variables for branch origin (A, B, C) were statistically essential. Omitting these highly significant variables would create severe **omitted variable bias**, which would contaminate the estimates of all other parameters, including the real estate variables. Their inclusion increased the Adjusted R^2 by approximately 5 percentage points, and their high significance confirms they are a critical part of the data's story.

The resulting model, using Bank B as the reference category, is:

$$\begin{aligned}\ln(\widehat{\text{OrdMargin}}) = & 1.557 + 0.239 \ln(\text{Commissions}) + 0.108 \ln(\text{SmallMed}) \\ & + 0.102 \ln(\text{Consumption}) + 0.108 \ln(\text{Deposits}) \\ & + 0.070 \ln(\text{Promoter}) - 0.022 \ln(\text{Mortgages}) \\ & - 0.334D_A + 0.284D_C\end{aligned}\tag{3}$$

4.2 Interpretation of the Real Estate Model

As anticipated, the forced inclusion of both collinear variables renders the $\text{LOG}(\text{MORTGAGES})$ coefficient statistically insignificant ($p = 0.5197$). Due to the multicollinearity, we cannot trust this specific estimate or claim that 'Mortgages' has any impact in *this specific model*.

However, the model *does* allow us to answer the director's question about $\text{LOG}(\text{PROMOTER})$'s effect:

- Holding all other variables constant, a **1% increase in Promoter loans is associated with a 0.070% increase** in the Ordinary Margin, and this effect is statistically significant ($p < 0.001$).

Crucially, this model also answers the question of whether differences exist between the banks. The p-values for both D_A and D_C are highly significant ($p < 0.0001$), proving that **there are significant structural differences** in profitability between the branch types. Holding all portfolio variables constant:

- Branches from Bank A (D_A) are, on average, 33.4% less profitable than branches from the reference (Bank B).
- Branches from Bank C (D_C) are, on average, 28.4% more profitable than branches from the reference (Bank B).

This model successfully fulfills the managerial request by quantifying the impact of **Promoter** while correctly controlling for structural differences between the banks.

5 Effect of Investment Funds on the Ordinary Margin

Finally, we address the managerial question regarding the impact of **Investment Funds** on the Ordinary Margin. The case requires us to fit a model that includes this variable and maximizes explanatory power.

5.1 Model Specification

Following the same robust methodology as the previous section, we began by forcing $\text{LOG}(\text{INVESTMENT FUNDS})$ into the model. We then performed a forward-stepwise procedure (maximizing Adjusted R^2) to add the most significant predictors.

Once again, the dummy variables D_A and D_C were included, as their high significance confirms that omitting them would bias our estimates. The resulting model, which isolates the effect of Investment Funds while controlling for all other major drivers, is:

$$\begin{aligned}
\ln(\widehat{\text{OrdMargin}}) = & 1.473 + 0.225 \ln(\text{Commissions}) + 0.107 \ln(\text{SmallMed}) \\
& + 0.068 \ln(\text{Promoter}) + 0.096 \ln(\text{Consumption}) \\
& + 0.106 \ln(\text{Deposits}) + 0.007 \ln(\text{Investment Funds}) \\
& - 0.333D_A + 0.281D_C
\end{aligned} \tag{4}$$

5.2 Interpretation of the Investment Funds Model

This model provides a clear and definitive answer. The coefficient for `LOG(INVESTMENT FUNDS)` is 0.00658.

This result is both **economically and statistically insignificant**.

- **Economically:** A coefficient of 0.0066 implies that a 1% increase in Investment Funds is associated with only a 0.0066% increase in the Ordinary Margin—an effect that is effectively zero.
- **Statistically:** This aligns with our findings from the "Optimal Model" selection (in Section section3), which also rejected `LOG(INVESTMENT FUNDS)` for having a high p-value ($p > 0.3$).

The conclusion is clear: Once we account for the main drivers of profit (like commissions, loans to SMEs, and deposits), Investment Funds do not provide any additional, significant explanatory power over the Ordinary Margin.

6 Methodology for Bonus Allocation

The final task is to propose a statistically sound methodology for identifying the 8 branches with the "best results" to award a bonus.

6.1 Critique of Naive Ranking

A "naive" approach would be to simply rank all 129 branches by their raw `Ordinary_Margin` and award the bonus to the top 8. This method is statistically flawed and managerially counterproductive.

A branch's high margin might not be due to the director's skill, but rather to circumstantial advantages:

- It might be a large branch with a high volume of 'Deposits' and 'Commissions'.
- It might be in a wealthy area with high 'SmallMed' loan demand.
- It might be one of the structurally different "Type C" branches.

This method simply rewards size or location, not *performance* or *efficiency*.

6.2 Proposed Methodology: Residual Analysis

The correct econometric approach is to use Residual Analysis. This method allows us to statistically isolate true managerial effectiveness.

6.2.1 Theoretical Explanation

Our "Optimal Model" (from Section 3) is our best-in-class tool for predicting what a branch's margin *should be*, given its specific characteristics (its volume of `Commissions`, `SmallMed` loans, `Promoter` loans, etc.).

For any branch i , we define its residual (ϵ_i) as the difference between its *actual* margin and its *predicted* margin:

$$\epsilon_i = Y_i - \hat{Y}_i \tag{5}$$

where:

- $Y_i = \ln(\text{Actual OrdMargin})$ is the real-world result.
- $\hat{Y}_i$ is the $\ln(\text{Predicted OrdMargin})$ given by our Optimal Model's equation.

The residual ϵ_i represents the portion of the Ordinary Margin that our model cannot explain. It is the statistical measure of unobserved factors, such as managerial skill, local efficiency, or superior customer service.

- A negative residual ($\epsilon_i < 0$) means the branch *under-performed* what was expected, given its portfolio and advantages.
- A residual near zero ($\epsilon_i \approx 0$) means the branch performed exactly as expected.
- A large, positive residual ($\epsilon_i > 0$) means the branch *significantly over-performed*. This branch's director achieved a higher margin than our model predicted, even after accounting for all their advantages. This is the statistical definition of efficiency.

6.2.2 Recommendation

To find the 8 directors with the "best results," the bank should:

1. Use our final Optimal Model from Section 3 (the 5-variable model).
2. Use this model to calculate the predicted $\ln(\widehat{\text{OrdMargin}})$ for all 129 branches.
3. Calculate the residual ϵ_i for all 129 branches.
4. Rank the branches from highest to lowest based on this residual.

The bonus must be awarded to the **8 branches with the highest positive residuals**. This ensures the bank rewards true managerial effectiveness and efficiency, not just circumstance.

7 Conclusion and Future Work

7.1 Conclusion

This report has successfully developed a robust econometric framework to explain the drivers of the Ordinary Margin in the bank's 129 branches. Our analysis began by addressing significant non-normality in the data through a log-log transformation.

We diagnosed and addressed the critical problem of multicollinearity, which prevented the use of a "full" model. By performing a best-subsets regression, we identified a parsimonious and powerful Optimal Model (Adjusted $R^2 = 84.23\%$) that identified **Commissions**, **SmallMed**, **Promoter**, **Consumption**, and **Deposits** as the core, significant drivers of profitability.

Furthermore, we addressed specific managerial requests by building constrained models.

- The Real Estate Model confirmed that **Promoter** loans have a significant positive impact, but the effect of **Mortgages** could not be statistically isolated due to collinearity.
- The Investment Funds Model definitively concluded that this variable has no significant economic or statistical impact on the Ordinary Margin once other primary drivers are accounted for.
- All models that included the structural dummy variables confirmed (with $p < 0.0001$) that there are systemic, significant differences in profitability between the branches of the acquiring bank (C) and the absorbed banks (A and B).

Finally, we proposed a methodologically superior approach for performance-based bonuses, recommending the use of Residual Analysis from our Optimal Model. This ensures the bank rewards true managerial efficiency rather than circumstantial advantages.

7.2 Limitations and Future Work

While our models are robust, this analysis can be extended and improved.

First, we must note a key constraint of our primary Optimal Model. It was developed **without** the structural dummy variables, as per the request to create a model based only on the continuous predictors. We must report that including these dummies (which were highly significant in all other models) would increase the Adjusted R-squared by approximately 4-5 percentage points, creating an even more powerful predictive model.

For future work, we recommend three paths to a more robust model:

1. **Robust Outlier Analysis:** Our current model is sensitive to the best and worst-performing branches. A more advanced technique would be to remove the 10% of branches with the most extreme residuals (both positive and negative), re-estimate the model on the central 90% to get a more stable "baseline", and **then** use this new model to analyze the residuals of the outliers. This would confirm if they are truly "star" performers or just a different statistical population.
2. **Feature Scaling:** Before applying more advanced models, all continuous variables should be standardized (e.g., scaled to Z-scores).
3. **Regularized Regression:** Once scaled, L1 (Lasso) and L2 (Ridge) regression techniques could be applied. These methods are specifically designed to handle collinearity by "shrinking" the coefficients of less-important variables, potentially yielding a more stable and predictive model than the best-subsets approach.